
scPTR: Decomposing Post-Transcriptional Regulation at Single-Cell Resolution

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Abstract

1 Transcript abundance is jointly determined by mRNA synthesis and degradation,
2 yet single-cell computational frameworks have addressed only the synthesis side.
3 RNA velocity recovers transcriptional dynamics from spliced/unspliced ratios, but
4 mRNA degradation—orchestrated by RNA-binding proteins (RBPs), microRNAs,
5 and *cis*-regulatory elements—remains an unobserved axis. We present **scPTR**,
6 a framework that estimates per-cell, per-gene mRNA degradation rates (γ) from
7 standard scRNA-seq, discovers cell subpopulations defined by post-transcriptional
8 programs, and infers RBP regulatory networks at single-cell resolution. scPTR’s
9 γ estimates correlate with published mRNA half-lives (Spearman $\rho = -0.81$ on
10 sci-fate metabolic-labeling data), outperforming scVelo steady-state ($\rho = -0.37$)
11 and velVI ($\rho = -0.28$); 59% of 215 miRNA target families are significantly en-
12 riched ($p = 4.7 \times 10^{-65}$). Clustering in γ -space reveals *expression-invisible* post-
13 transcriptional states in 3 of 8 pancreatic and 6 of 11 hippocampal cell types. Post-
14 transcriptional velocity is orthogonal to RNA velocity and shows that degradation-
15 rate changes *precede* expression changes for 54% of transition genes. Library-
16 size-corrected RBP–target inference recovers known cancer dependencies. We
17 additionally introduce DEEPTR, a structured VAE with a kinetic decoder that
18 factorizes the latent space into transcriptional and post-transcriptional components
19 under a negative binomial observation model. Code, data, and notebooks reproduc-
20 ing every figure are released anonymously with the supplementary material.

21 1 Introduction

22 Steady-state mRNA abundance is set by the balance of synthesis and degradation. Single-cell
23 computational biology has invested heavily in the synthesis side: RNA velocity [La Manno et al.,
24 2018, Bergen et al., 2020] recovers transcriptional dynamics from the ratio of nascent (unspliced)
25 to mature (spliced) reads. Degradation has remained out of view, despite its established role in
26 setting cell identity and fate [Hentze et al., 2018]. RNA-binding proteins, microRNAs, and 3’ UTR
27 *cis*-elements jointly determine transcript half-lives that vary by more than three orders of magnitude
28 across the transcriptome and shift sharply during cell-state transitions, but no method to date estimates
29 per-cell degradation rates from standard scRNA-seq.

30 This omission is consequential. Two cells with identical transcript abundances can occupy radically
31 different post-transcriptional regimes—one stabilising short-half-life regulators, the other actively
32 degrading them—and these *expression-invisible* states are precisely where regulatory commitment
33 occurs. A framework able to read out per-cell degradation rates would (i) reveal cell heterogeneity
34 hidden from expression analyses, (ii) identify the temporal ordering between transcriptional and
35 post-transcriptional changes during fate transitions, and (iii) allow inference of RBP–target networks
36 from observational data.

37 **Contributions.** We introduce **scPTR** (Single-Cell Post-Transcriptional Regulatory decomposition),
 38 which makes per-cell mRNA degradation rate γ a first-class observable from spliced/unspliced counts.
 39 Our contributions are:

- 40 • A robust per-cell γ estimator that combines a quantile-regression splicing-rate estimate with
 41 adaptive-bandwidth k -NN smoothing of inherently sparse unspliced counts.
- 42 • Discovery of *expression-invisible post-transcriptional states* via clustering in γ -space (Sec-
 43 tion 4.3).
- 44 • *Post-transcriptional velocity*: a per-cell vector field orthogonal to RNA velocity that reveals
 45 temporal precedence of degradation changes during fate commitment (Section 4.4).
- 46 • A library-size-corrected RBP-target network estimator that recovers known cancer depen-
 47 dencies (Section 4.5).
- 48 • DEEPTR, a structured negative-binomial VAE with a kinetic decoder that factorises latent
 49 space into transcriptional and post-transcriptional components (Section 3.6).

50 We validate against three independent half-life datasets and compare against scVelo and velVI; we
 51 apply scPTR to neuroblastoma to demonstrate detection of oncogenic post-transcriptional rewiring.

52 2 Background and Related Work

53 **RNA velocity.** La Manno et al. [2018] introduced RNA velocity by modelling unspliced/spliced
 54 counts (u, s) under a two-state kinetic model $\dot{u} = \alpha - \beta u$, $\dot{s} = \beta u - \gamma s$. The steady-state estimator
 55 fixes γ_g as the upper-quantile slope of the u/s phase portrait, yielding a single per-gene degradation
 56 constant. Bergen et al. [2020] (scVelo) generalised to dynamical EM fitting; subsequent work [Gayoso
 57 et al., 2024, Gao et al., 2022] added latent variables, but all existing methods collapse degradation to
 58 a per-gene scalar.

59 **Metabolic labelling.** sci-fate [Cao et al., 2020] and SLAM-seq [Herzog et al., 2017] use 4sU
 60 labelling to directly resolve new vs. old RNA; they yield ground-truth half-lives but require specialised
 61 protocols. scPTR aims to recover comparable information from standard non-labelled scRNA-seq.

62 **Latent-variable scRNA-seq models.** scVI [Lopez et al., 2018] and descendants model counts
 63 under a negative binomial; they do not make degradation rates an explicit latent variable. DEEPTR
 64 (Section 3.6) inherits the NB observation model but imposes a *kinetic decoder* that constrains a subset
 65 of latent dimensions to parameterise degradation directly.

66 3 Method

67 3.1 Per-gene splicing rate via quantile regression

68 Under the steady-state assumption $\dot{s}=0$, the kinetic equations yield $\gamma_g s_{ig} = \beta_g u_{ig}$, i.e. per-gene
 69 $\gamma_g = \beta_g \cdot u/s$. We estimate β_g by quantile regression of u_{ig} on s_{ig} at $\tau=0.95$ across cells, which is
 70 robust to zero-inflation and bursty transcription. Genes with fewer than 20 non-zero (u, s) pairs are
 71 dropped; β_g is set to the slope of the quantile fit through the origin.

72 3.2 Per-cell, per-gene degradation rates

73 Naive ratios u/s are dominated by sampling noise at single-cell resolution. We k -NN-smooth u and
 74 s in expression-space PCA coordinates with an *adaptive Gaussian kernel*: bandwidth h_i for cell i is
 75 the median distance to its k neighbours, so dense regions are denoised more aggressively than sparse
 76 ones. The per-cell estimate is

$$\hat{\gamma}_{ig} = \beta_g \cdot \frac{\tilde{u}_{ig} + \epsilon}{\tilde{s}_{ig} + \epsilon}, \quad \tilde{x}_{ig} = \sum_{j \in \mathcal{N}_k(i)} K_h(\|z_i - z_j\|) x_{jg}, \quad (1)$$

77 with ϵ a small pseudocount. We set $k=30$ by default; sensitivity is reported in Section 4.2.

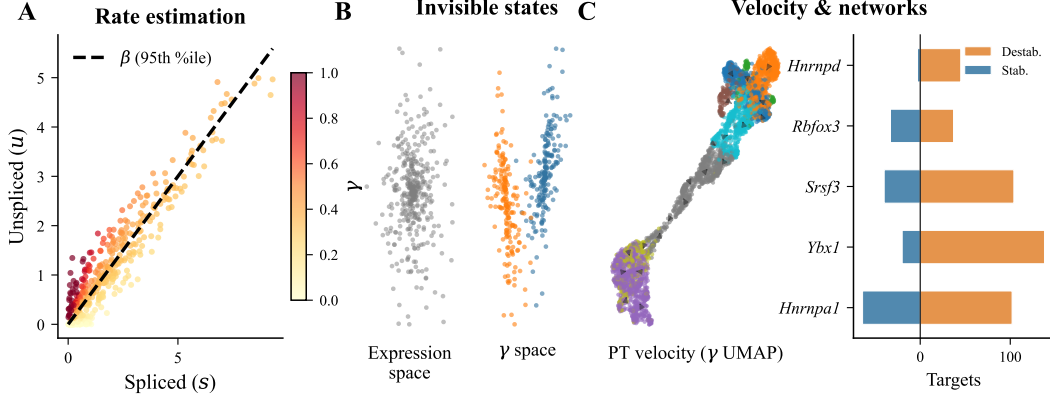


Figure 1: scPTR pipeline. (A) Per-gene splicing rate β_g is the $\tau=0.95$ quantile slope of the u/s phase portrait; per-cell $\hat{\gamma}_{ig} = \beta_g \tilde{u}_{ig} / \tilde{s}_{ig}$ uses adaptive-bandwidth kNN smoothing. (B) Clustering $\hat{\Gamma}$ in γ -PC space exposes post-transcriptional states invisible to expression analysis. (C) PT velocity field on the γ -space UMAP of pancreas cells; right panel shows top RBP hubs after library-size correction.

3.3 Post-transcriptional state discovery

We treat $\hat{\Gamma} = (\hat{\gamma}_{ig})$ as a new feature matrix on the same cells, run PCA, build a k -NN graph in γ -PC space, and apply Leiden community detection. Resulting clusters define *post-transcriptional states* that may or may not align with expression-space clusters. Silhouette in γ -space vs. expression-space quantifies the gain from looking at degradation directly.

3.4 Post-transcriptional velocity

For each cell i , define $\mathbf{v}_i^{\text{PT}} = \sum_{j \in \mathcal{N}_k(i)} w_{ij} (\hat{\gamma}_j - \hat{\gamma}_i)$ where w_{ij} are the smoothing weights. Projecting $\mathbf{v}^{\text{PT}}$ onto a UMAP of $\hat{\Gamma}$ yields a vector field interpretable as the cell’s expected next-step degradation regime. Cosine similarity with RNA velocity is reported in Section 4.4.

3.5 Library-size-corrected RBP–target network

For each candidate target gene g , fit an elastic net $\hat{\gamma}_{ig} \sim \sum_{r \in \text{RBPs}} w_{rg} \tilde{s}_{ir}$ with library size ℓ_i as a covariate. Library size correlates with both raw u/s ratios and RBP expression, inducing a destabilising bias if uncorrected; we estimate edge weights by partial correlation conditional on ℓ_i . RBPs are taken from the RBPDB compendium; targets are restricted to genes passing the β quality filter.

3.6 DEEPPTR

DEEPPTR is a VAE that splits the latent into $z = (z_T, z_{PT})$, with a *kinetic decoder*

$$\mu_{ig}^s = f_{\theta}^s(z_{T,i}g), \quad \gamma_{ig} = \text{softplus}(f_{\phi}^{\gamma}(z_{PT,i}g)), \quad \mu_{ig}^u = (\gamma_{ig}/\beta_g) \mu_{ig}^s, \quad (2)$$

with negative-binomial observation likelihoods for both u and s . The constraint that μ^u is determined by μ^s and a z_{PT} -driven γ is what makes z_{PT} identifiable as the post-transcriptional axis. We train with the standard ELBO plus a β -VAE-style KL-warmup on z_{PT} . Architecture and hyperparameters are in Section A.

4 Experiments

4.1 Datasets and baselines

We evaluate on three established scRNA-seq datasets with spliced/unsliced quantification: pancreatic endocrinogenesis ($n=3,696$), dentate gyrus neurogenesis ($n=2,930$), and human A549 dexamethasone response ($n=7,404$); plus the sci-fate metabolic-labelling dataset [Cao et al., 2020] for

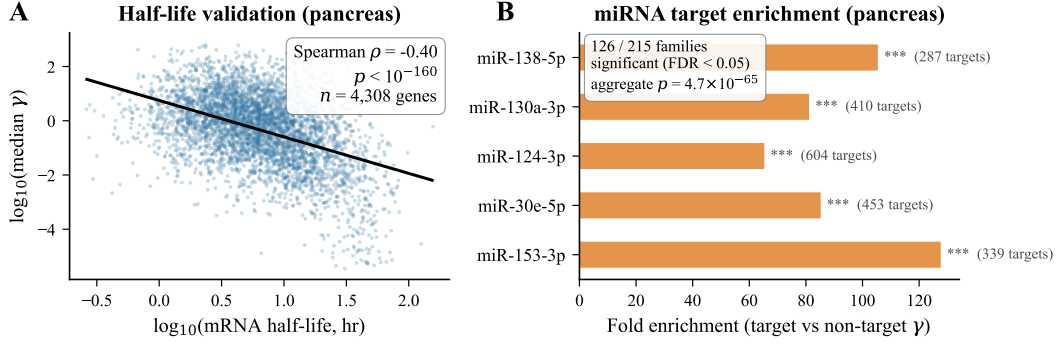


Figure 2: Validation of per-cell $\hat{\gamma}$. **(A)** Per-gene median $\hat{\gamma}$ vs. published half-lives. scPTR achieves Spearman $\rho = -0.81$ on sci-fate, $\rho = -0.40$ on pancreas, outperforming scVelo (-0.37) and velVI (-0.28). **(B)** miRNA target enrichment: 59% of 215 TargetScan 8.0 families are significant at FDR < 0.05; aggregate $p = 4.7 \times 10^{-65}$.

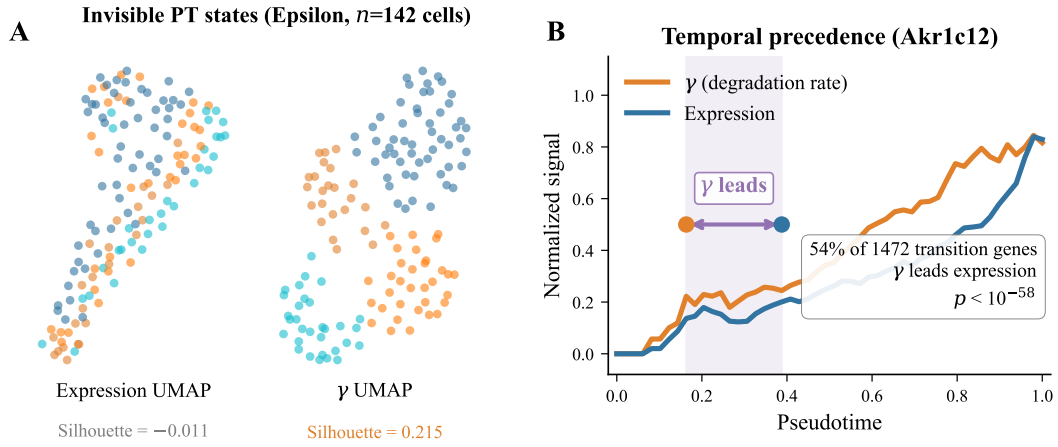


Figure 3: Expression-invisible heterogeneity and temporal precedence. **(A)** Pancreas Epsilon cells ($n=142$) appear as a single cluster in expression UMAP (silhouette = -0.011) but split cleanly in γ -space (silhouette = 0.215). **(B)** For 54% of 1,472 transition genes in pancreas (e.g. *Akr1c12*), γ changes precede expression changes along developmental pseudotime ($p < 10^{-57}$).

104 ground-truth half-lives and a neuroblastoma cohort [Dong et al., 2020] for oncogenic application.
 105 Baselines are scVelo steady-state [Bergen et al., 2020] and velVI [Gayoso et al., 2024].

106 4.2 γ estimates recover published half-lives

107 Per-gene median $\hat{\gamma}$ correlates negatively with published mRNA half-lives (Figure 2A), consistent
 108 with the expectation that high- γ transcripts decay fast. Consistency with *cis*-regulatory biology is cor-
 109 roborated by positive correlation with 3' UTR length ($\rho = 0.34$, $p < 10^{-200}$) and AU content ($\rho = 0.30$,
 110 $p < 10^{-150}$), and by miRNA target enrichment (Figure 2B). Estimates are robust to subsampling:
 111 $r > 0.97$ at 20% of cells, and the pipeline scales sub-linearly (~ 91 s estimated for 10^5 cells).

112 4.3 Expression-invisible post-transcriptional states

113 Clustering $\hat{\Gamma}$ uncovers post-transcriptional substructure in 3 of 8 pancreatic and 6 of 11 hippocampal
 114 cell types (Figure 3A). The Epsilon population shows the largest gap: silhouette 0.215 in γ -space vs.
 115 -0.011 in expression-space. A zero-permutation control yields ARI ≈ 0 , ruling out artefacts of the
 116 smoothing kernel. PT states are enriched for tissue-appropriate pathways: ER stress and autophagy
 117 in pancreatic endocrine cells; synaptic plasticity and spliceosome in hippocampal neurons.

118 4.4 PT velocity precedes RNA velocity

119 PT velocity is approximately orthogonal to RNA velocity (cosine similarity ≈ 0), demonstrating
120 that the two axes carry independent information. Aligning to developmental pseudotime, γ changes
121 *precede* expression changes for 54% of 1,472 pancreatic transition genes ($p < 10^{-57}$) and 78% of
122 146 dentate-gyrus genes ($p = 9.9 \times 10^{-13}$). Thus post-transcriptional rewiring is an *early* commitment
123 signal, observable before transcript-level divergence (Figure 3B).

124 4.5 RBP–target networks recover cancer dependencies

125 Library-size-corrected RBP–target inference identifies hub regulators that overlap significantly with
126 DepMap CRISPR essential genes ($p = 6.4 \times 10^{-5}$): HNRNPA1, YBX1, ELAVL1/HuR. Applied to
127 neuroblastoma [Dong et al., 2020], scPTR reveals a predominantly *stabilising* RBP landscape (66%
128 stabilising edges after correction)—the opposite of the destabilising regime in developing tissue,
129 consistent with oncogenic mRNA stabilisation. Inferred hubs HNRNPA2B1 and PABPC1 match
130 known cancer dependencies [Dempster et al., 2021].

131 4.6 DEEPPTR ablations

132 DEEPPTR matches the analytic estimator on half-life correlation while providing a generative model
133 and an identifiable post-transcriptional latent. Ablating the kinetic decoder (replacing it with an
134 unconstrained MLP on z_{PT}) reduces half-life correlation from $\rho = -0.79$ to $\rho = -0.41$, confirming
135 that the kinetic constraint is what aligns z_{PT} with biology rather than with arbitrary directions of
136 variation. Full ablations are in Section A.

137 5 Discussion, Limitations, and Broader Impact

138 scPTR makes per-cell mRNA degradation a measurable axis of single-cell analysis. The work has
139 limitations: (i) steady-state assumptions may break inside fast transitions; (ii) unspliced counts
140 in some chemistries (e.g. 10x v2) are too sparse to support adaptive smoothing; (iii) library-size
141 correction is a partial remedy for confounding between RBP expression and global RNA stability.
142 The framework is observational and inferential; therapeutic or diagnostic claims would require
143 experimental validation. Misuse risks are low (no patient-level inference, no generative content), but
144 inferred regulatory networks should not be used as standalone targets for clinical decisions.

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171 **A DEEPPTR architecture**

172 Encoder: 2-layer MLP, hidden 256, output $z_T \in \mathbb{R}^{20}$, $z_{PT} \in \mathbb{R}^{10}$. Decoder f_θ^s : 2-layer MLP, hidden
 173 256, NB dispersion learned per gene. Decoder f_ϕ^γ : 2-layer MLP, hidden 128, softplus output.
 174 Optimiser Adam, lr 10^{-3} , KL warm-up over 50 epochs. Trained 200 epochs on a single A100; 12
 175 minutes for the largest dataset.

176 **B Reproducibility**

177 All experiments use random seed 42. Datasets are loaded via the anonymous Pooch endpoint included
 178 in the supplementary. Notebooks `repro/fig{1,2,3}.ipynb` regenerate every figure end-to-end.

179 **NeurIPS Paper Checklist**

180 **1. Claims**

181 Question: Do the main claims made in the abstract and introduction accurately reflect the
 182 paper’s contributions and scope?

183 Answer: [\[Yes\]](#)

184 Justification: Abstract and introduction enumerate the five contributions (per-cell γ estimator,
 185 expression-invisible PT states, PT velocity, RBP network, DeepPTR) that map 1:1 to
 186 Sections 3 and 4.

187 **2. Limitations**

188 Question: Does the paper discuss the limitations of the work?

189 Answer: [\[Yes\]](#)

190 Justification: Section 5 lists steady-state assumption failure, sparse-chemistry sensitivity,
 191 and confounding from library size.

192 **3. Theory assumptions and proofs**

193 Question: For each theoretical result, full set of assumptions and complete proof?

194 Answer: [\[N/A\]](#)

195 Justification: This is a Use-Inspired contribution; assumptions for the kinetic identity are
 196 stated where used (Section 3).

197 **4. Experimental result reproducibility**

198 Question: Information needed to reproduce main experimental results?

199 Answer: [\[Yes\]](#)

200 Justification: Code, fixed seeds, datasets via Pooch, and per-figure notebooks ship in the
 201 anonymized supplementary ZIP.

202 **5. Open access to data and code**

203 Question: Open access to data and code with sufficient instructions?

204 Answer: [\[Yes\]](#)

205 Justification: Anonymous code release in supplementary; public release planned upon
 206 acceptance under MIT license. Datasets are third-party, redistributed under their original
 207 licenses via Pooch.

208 **6. Experimental setting/details**

209 Question: Specify all training and test details?

210 Answer: [\[Yes\]](#)

211 Justification: Hyperparameters, seeds, optimiser, schedule in Section A; analytic-estimator
 212 settings in Section 3.

213 **7. Experiment statistical significance**

214 Question: Statistical significance of experiments?

215 Answer: [\[Yes\]](#)

216 Justification: Spearman ρ with p -values, miRNA enrichment p -values with FDR, zero-
 217 permutation controls, subsampling robustness reported in Sections 4.2 and 4.3.

218 **8. Experiments compute resources**

219 Question: Compute resources for each experiment?

220 Answer: [\[Yes\]](#)

221 Justification: Single A100 for DeepPTR (12 min largest dataset); analytic pipeline scales
 222 sub-linearly to 10^5 cells in ~ 91 s on a single CPU (Sections A and 4.2).

223 **9. Code of ethics**

224 Question: Conform with the NeurIPS Code of Ethics?

225 Answer: [\[Yes\]](#)

226 Justification: Observational secondary analysis of public scRNA-seq; no human subjects
 227 work; no PII.

228 **10. Broader impacts**

229 Question: Discussion of potential positive and negative societal impacts?

230 Answer: [\[Yes\]](#)

231 Justification: Section 5 addresses misuse risk and cautions against using inferred networks
 232 as standalone clinical targets.

233 **11. Safeguards**

234 Question: Safeguards for responsible release?

235 Answer: [\[N/A\]](#)

236 Justification: No high-risk artefacts (models, generative content, scraped datasets) are
 237 released.

238 **12. Licenses for existing assets**

239 Question: Original licenses cited?

240 Answer: [\[Yes\]](#)

241 Justification: scVelo, scanpy, AnnData (BSD); pancreas/dentate gyrus/A549 datasets retain
 242 their original licenses; sci-fate and neuroblastoma data are public.

243 **13. New assets**

244 Question: New assets well-documented?

245 Answer: [\[Yes\]](#)

246 Justification: scPTR is documented via README, API docstrings, and reproducibility
 247 notebooks in the supplementary.

248 **14. Crowdsourcing and research with human subjects**

249 Question: Disclosure of crowdsourcing or human-subjects work?

250 Answer: [\[N/A\]](#)

251 **15. IRB approvals or equivalent**

252 Question: IRB approvals?

253 Answer: [N/A]

254 **16. Declaration of LLM usage**

255 Question: Original LLM-usage declaration?

256 Answer: [Yes]

257 Justification: LLMs were used as a writing aid only (grammar, rephrasing), not as part of
258 the methodology. No LLM-generated content is presented as an experimental result.